

PROBLEM SET 3

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions: Due on 17/09/2021; scan or type your work and submit through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

1. Consider the set of lotteries on the set of monetary prizes $Z = \{0, 1, 5\}$, this set $\mathcal{L}(Z)$ can be identified with the three-dimensional unit simplex, i.e., $\mathcal{L}(Z) = \Delta(Z) = \{(p_1, p_2, p_3) \in \mathbb{R}^3 | 0 \leq p_i \leq 1, i = 1, 2, 3, \text{ and } p_1 + p_2 + p_3 = 1\}$.
 - (a) Consider the following three lotteries on $\Delta(Z)$: $(1/3, 1/3, 1/3)$, $(1/4, 1/4, 1/2)$, $(1/5, 3/5, 1/5)$. Does any lottery first-order stochastically dominate another?
 - (b) For any $p \in \Delta(Z)$, let $\succsim_{\text{fosd}}(p) \subseteq \Delta(Z)$ be the set of lotteries that first-order stochastically dominate p . Show that $\succsim_{\text{fosd}}(p)$ is convex and closed.
 - (c) The *Marshack-Machina Triangle*, Δ_M , is a useful graphical apparatus to analyze graphically the set of lotteries over a three-element prize set. It is the projection of the simplex $\Delta(Z)$ to the two-dimensional Euclidean space and is defined as follows:

$$\begin{aligned}\Delta_M &= \{(p_1, p_3) \in \mathbb{R}^2 | (p_1, p_2, p_3) \in \Delta(Z)\} \\ &= \{(p_1, p_3) \in \mathbb{R}^2 | 0 \leq p_1 \leq 1, 0 \leq p_3 \leq 1, 0 \leq p_1 + p_3 \leq 1\}\end{aligned}$$

Make a diagram of Δ_M and depict in your diagram the three lotteries in part (a).

- (d) Which area of the Marshack-Machina triangle corresponds to $\succsim_{\text{fosd}}(p)$ for an arbitrary lottery p ?
- (e) An iso-expected value line for an arbitrary lottery $p \in \Delta(Z)$ is the set of all lotteries which have the same expected value as p , i.e., $\{q \in \Delta(Z) | E[p] = E[q]\}$. How does the iso-expected value line look like in the Marshack Machina triangle? Draw the iso-expected lines for the lotteries in part (a).
- (f) For any $p \in \Delta(Z)$, let $\succsim_{\text{sosd}}(p) \subset \Delta(Z)$ be the set of lotteries that second-order stochastically dominate p . Do the same exercises as in (b) and (d) with respect to $\succsim_{\text{sosd}}(p)$. *Hint: the result in part (e) will be helpful.*
- (g) Consider any Bernoulli utility function $u : Z \rightarrow \mathbb{R}$, which is increasing with respect money z , and suppose the decision maker evaluates lotteries according to their expected utility $E[u(p)] = \sum_{z \in Z} p(z)u(z) = p_1u(0) + p_2u(1) + p_3u(5)$. Recall that an indifference curve of lottery p is the set of all lotteries for which the expected utility is the same, i.e., $\{q \in \Delta(Z) | E[u(q)] = E[u(p)]\}$. How does the indifference curve of an arbitrary lottery p looks like in the Marshack-Machina triangle? Draw the indifference curves for the lotteries in part (a) in the Marshack-Machina triangle for an arbitrary increasing Bernoulli utility function u .
- (h) Let $p \in \Delta(Z)$ be an arbitrary lottery. Using what you learned in part (g), how does the indifference curve for lottery p compare to the iso-expected value line for p in the Marshack-Machina triangle when:
 - i. u exhibits risk aversion?
 - ii. u exhibits risk loving?
 - iii. u exhibits risk neutrality?
- (i) Suppose that u_1 and u_2 are two strictly increasing Bernoulli utility functions over money, and suppose further that u_1 is more risk averse than u_2 . Let $p \in \Delta(Z)$ be an arbitrary lottery. How does the indifference curve at p for u_1 compare to the indifference curve at p for u_2 ?

2. Suppose a decision maker lives for two periods, t_0 (today) and t_1 (tomorrow). In period t_1 there are three possible outcomes:
- State s_1 with probability 0.3 (recession).
 - State s_2 with probability 0.4 (normal growth).
 - State s_3 with probability 0.3 (expansion).

There are two stocks available to the decision maker, α and β . The stocks yield payoffs according to the following schedule:

Stock	s_1	s_2	s_3
α	7	12	14
β	15	5	10

- If possible, rank the stocks by the criterion of first order stochastic dominance. *Hint: you can use a graph to answer this question.*
- If possible, rank the stocks by the criterion of second order stochastic dominance. *Hint: if you made an useful graph from part (a), then you will have a hint about how to proceed.*
- Consider an investor with Bernoulli utility function $u(w) = \sqrt{w}$. After the investor spends what is needed for consumption at t_0 , the investor ends up with current wealth of w plus one share of stock α . The investor can either keep the share until next period or sell it now. For simplicity, assume that savings earn a risk free interest rate of zero.
 - Suppose $w = 10$. What is the minimum price the investor would be willing to sell the stock for? Is this more or less than the expected payoff from α ? Why?
 - Suppose now that $w = 10,000$. What is the minimum price the investor would be willing to sell the stock for? How do you explain the difference between this answer and your answer under the previous part? *Hint: compute, for any level of wealth, the coefficient of absolute risk aversion of $u(w) = \sqrt{w}$. How does it behave for increasing levels of wealth? Then get inspired by the solutions from problem set 2, (the insurance problem). What do you conclude?*